AGH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Power Series and Taylor Series for Electronics and Telecommunications

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March 18, 2022

Function space as a domain of convergence

Consider a sequence of functions $f_n : [a,b] \to \mathbb{R}$.

The sequence (f_n) is pointwise convergent $f_n \xrightarrow[n\to\infty]{} f$ iff

$$\forall \varepsilon > 0 \, \forall x \in [a,b] \, \exists N = N(x,\varepsilon) \, \forall n \geqslant \mathbb{N} : \, |f_n(x) - f(x)| < \varepsilon$$

The sequence (f_n) is uniformly convergent $f_n \overset{}{\underset{n\to\infty}{\rightrightarrows}} f$ iff

$$\forall \varepsilon > 0 \, \exists N = N(\varepsilon) \, \forall x \in [a,b] \, \forall n \geqslant \mathbb{N} : \, |f_n(x) - f(x)| < \varepsilon$$

But there are many other ways to define the limit in a space of functions. In some sens those are the two extremal ways: very weak (pointwise), and very strong (uniform). Weak requirements would produce weak consequences, but strong requirements would be very hard to satisfy.

Function series $\sum_{n=1}^{\infty} f_n(x)$

Let (f_n), $n \in \mathbb{N}$ be a sequence of functions, which are defined at the same domain $x \in [a, b]$. Let's define partial sums of the function sequence

$$s_n = f_1 + f_2 + \ldots + f_n, \text{ for any } n \in \mathbb{N}.$$

The limit of the sequence of partial sums (if it actually exists) is called the sum of the series and is denoted as follows

$$\lim_{n\to\infty} s_n(x) = \sum_{n=1}^{\infty} f_n(x), \ x \in [a, b].$$

If the sequence of the partial sums is convergent/divergent, then we say that series is convergent/divergent, respectively.

Important function series

Power series are important, because Taylor series is a special type of a power series.
Notice that power series is a different notion, then a Taylor formula from Calculus. However, if investigated by the computer simulations obviously the difference between sometimes not appreciated by engineers. In Taylor formula we deal with an error estimation, with Taylor series is an "‘ideal"’ (ie. the limit) never acheaved in the simulations.

Power series

Power series

Power series is a function series of the form

$$\sum a_n(x-x_0)^n,\ a_n \in \mathbb{R},$$

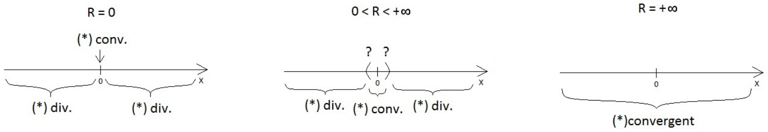
where x_0- is a center and often equal to 0, x - a real variable.

One can consider x being a complex variable, then we move to the area which is called complex analysis, and complex analytic functions.

Power series $\sum a_n x^n$

Denote by $Z := \{|x| : x \in \mathbb{R}$ st. $\sum a_n x^n$ is convergent$\}$. By $R = \sup Z$ we define the radius of convergence of the $\sum a_n x^n$.

Notice that $0 \in Z$, therefore $Z \neq \emptyset$, which implies that $0 \leq \sup Z \leq +\infty$. What are the possible cases for power series centered at 0?



One can easily extend it to the case $x_0 \neq 0$.

Theorem about convergence of a power series

The power series $\sum a_n x^n$ is absolutely and uniformly convergent on any closed interval $< -r, r >$ included in the interior of the convergence interval (ie. $(-R; R)$).

Does it mean that $\sum a_n x^n$ is uniformly convergent in the whole interval?
No, if the series is not convergent at R or $-R$.
This statement can be easily proven by the comparison test and upper bound being a geometric series.

Taylor series

Taylor series

For a function of class $\mathcal{C}^\infty$ Taylor series is a power series of the form

$$\sum_{n=0}^{+\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n,$$

where x_0- is a center of the series and when equal to 0 we call it Maclaurin series,
x - a real variable.

If the Taylor series is convergent to the function f in the entire domain of the function, then this function is called analytic.